FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Obstructed Labour, Malposition & Malpresentation

Tutorial | Week 6 · Obstruction, HIV, diabetes & the written test | Obstetrics | Dr Mutamba

Obstructed labour means the presenting part cannot descend despite good contractions. It is a preventable killer: it ruptures the uterus, causes sepsis and fistula and kills babies. Recognise it early (moulding 3+, caput, Bandl's ring, haematuria, a distended bladder) and treat it with resuscitation, a catheter, antibiotics and prompt caesarean section, never oxytocin. Malpositions (OP, deep transverse arrest) and malpresentations (breech, face, brow, shoulder, compound) are its commonest mechanical causes, and cord prolapse and shoulder dystocia are the emergencies that come with them.

Must know before the ward round

1. Obstructed labour = no descent of the presenting part despite strong contractions because of a mechanical block. Oxytocin is absolutely contraindicated.
2. Hallmarks are moulding 3+ (overlapping sutures that do not reduce), big caput, Bandl's ring, oedematous cervix or vulva, haematuria, a distended bladder, maternal exhaustion and fetal distress or death.
3. Resuscitate (IV Ringer's lactate or normal saline, at least 1.5 L), catheterise, give IV antibiotics, group and cross-match, then deliver within about 2 hours, usually by CS.
4. If labour was prolonged or obstructed, leave a Foley catheter in for 7–14 days to prevent or treat a fresh vesicovaginal fistula.
5. Brow presentation (mentovertical diameter 13.5 cm) and mentoposterior face cannot deliver vaginally at term. Mentoanterior face can.
6. Breech: offer ECV from 36 weeks (nulliparous) or 37 weeks (multiparous). Assisted breech means hands off until the umbilicus, then Lovset for the arms and Mauriceau–Smellie–Veit or Burns–Marshall for the head.
7. Cord prolapse: do not handle the cord, keep the presenting part up (manually, knee–chest, or bladder filled with 500–750 mL saline), give terbutaline 0.25 mg SC and do a category 1 CS.
8. Shoulder dystocia (HELPERR): call for help, McRoberts plus suprapubic pressure first, then internal manoeuvres or delivery of the posterior arm. Never use fundal pressure.
9. Mulago standards for every obstructed labour: IV fluids, IV antibiotics, vital-sign monitoring, catheter, group and cross-match, and delivery within 2 hours of diagnosis.
10. Always pair the vaginal station with the fifths palpable: 3/5 or more palpable in the second stage means not engaged, whatever the VE says.
11. Adequate contractions are 3 or more in 10 minutes lasting over 40 seconds; weak contractions with no signs of obstruction suggest inefficient uterine action, not obstruction.
12. On VE the posterior fontanelle is a triangle with 3 sutures (occiput) and the anterior fontanelle a diamond with 4 sutures; an easily felt anterior fontanelle means a deflexed head.

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1. Definition

- **Obstructed labour** happens when the presenting part cannot advance through the birth canal **despite adequate uterine contractions**, because the passage is too small for the passenger, or the passenger is presenting the wrong diameter.
- It is one of the "big five" killers in Uganda. It leads directly to **uterine rupture, PPH, puerperal sepsis, obstetric fistula** and **stillbirth or birth asphyxia**. It is also almost entirely **preventable** with good labour monitoring (the partograph or WHO Labour Care Guide) and timely caesarean section.

Term	Definition
Prolonged labour	Active first stage lasting longer than expected. On the WHO Labour Care Guide (LCG) an alert is triggered when the cervix does not progress within the time limits for each centimetre from 5 cm. The classic partograph uses the **alert line** (1 cm/hour) and the **action line** 4 hours to its right. Total labour > 24 hours is a traditional cut-off
Obstructed labour	Failure of descent of the presenting part **despite good contractions**, because of mechanical disproportion between the fetus and the birth canal
Cephalopelvic disproportion (CPD)	The fetal head is too big for that pelvis (absolute), or presents a diameter that is too big (relative, e.g. deflexed OP or brow)
Malposition	The vertex presents but the denominator (the occiput) is **not anterior**: occipito-posterior (OP) or occipito-transverse (OT)
Deep transverse arrest (DTA)	The head is in OT position **at or below the ischial spines** with a fully dilated cervix, and it fails to rotate or descend despite good contractions
Malpresentation	A presenting part other than the vertex: **breech, face, brow, shoulder (transverse or oblique lie) or compound**
Cord presentation / prolapse	The cord lies below the presenting part with **intact** membranes (presentation) or **ruptured** membranes (prolapse, which can be occult or overt)
Shoulder dystocia	A vaginal cephalic birth that needs **additional manoeuvres** to deliver the shoulders after routine gentle traction has failed (RCOG GTG 42)

CLINICAL PEARL

Never forget: a woman who has laboured long with a **hot, dry, tachycardic body, a tender lower segment, a rising Bandl's ring and bloody urine** is about to rupture her uterus. She needs **fluids, a catheter, antibiotics and theatre now**. She must **never** get oxytocin.

2. Epidemiology

- Obstructed labour complicates an estimated **~4–5% of births worldwide** and causes about **8% of maternal deaths**. Ugandan hospital studies report a much higher proportion of admissions (around **10%**) because

referral hospitals receive the neglected cases.

- At Mulago, obstructed labour and its complications (rupture, sepsis, PPH) accounted for about **a quarter of maternal deaths** in an audit. Obstructed labour caused about **73% of uterine ruptures** in Ugandan series, and uterine rupture alone caused roughly **1 in 10 maternal deaths** at the national referral hospital.
- The Mulago criteria-based audit set **six standards** for every woman with obstructed labour: **IV fluids, IV antibiotics, monitoring of maternal vital signs, bladder catheterisation, blood grouping and cross-matching, and delivery within 2 hours of diagnosis**.
- **Breech** occurs in about **20% of pregnancies at 28 weeks** but only **3–4% at term**, because most babies turn spontaneously.
- **Occipito-posterior** position occurs in 10–20% of labours at onset, but most rotate anteriorly. Only about **5% persist as OP** at delivery.
- **Face** presentation occurs in about 1 in 500 births and **brow** in about 1 in 1,000–1,500. **Transverse lie** at term occurs in about 1 in 300, and is much commoner in grand multiparas.
- **Cord prolapse** occurs in about 0.1–0.6% of births. **Shoulder dystocia** occurs in about 0.6–0.7% of vaginal cephalic births (RCOG).

> **UGANDA CONTEXT**
> The Ministry of Health is replacing the classic partograph with the **WHO Labour Care Guide**. Whichever tool your ward uses, the principle is the same: **plot, recognise deviation early, act**. At an **HC III** (no theatre), recognising prolonged labour means **referring early**, not "trying a little oxytocin". At an **HC IV or hospital** it means a decision for CS. Most obstructed labours seen at Mengo or Mulago are late referrals, and the delay happened before the woman arrived.

3. Aetiology and Risk Factors

The "3 Ps"

- **Passage**: a contracted pelvis (childhood malnutrition, adolescence, rickets), tumours.
- **Passenger**: macrosomia, hydrocephalus, **malposition**, **malpresentation**, locked twins.
- **Powers**: usually *strong* in obstruction. That is what separates obstruction from simple slow labour caused by poor contractions.

Risk factors in Uganda: adolescent pregnancy (a small, immature pelvis), short maternal stature (less than about 150 cm), childhood malnutrition, primigravidity, macrosomia (diabetes, post-dates), malpresentation, **delay in reaching care** (the three delays: deciding to seek care, reaching a facility, receiving care there), long labour at home or with traditional birth attendants, and HC IIIs without theatre capacity.

Why malpresentations happen. Anything that lets the baby move freely, or stops the head settling into the pelvis:

- **Lax uterus**: grand multiparity.
- **Excess room**: polyhydramnios, or a preterm baby.
- **Something blocking the pelvis**: placenta praevia, fibroids, a contracted pelvis, ovarian tumours.
- **Abnormal shape**: uterine anomalies (bicornuate, septate) or twins.
- **Fetal factors**: anencephaly, hydrocephalus, neck masses (these cause face presentation), neuromuscular problems and fetal death.

> **CLINICAL PEARL**
> When you find a malpresentation, ask **"why?"** Scan for **placenta praevia** before any vaginal examination or ECV, and look for fetal anomalies, polyhydramnios, fibroids or a uterine anomaly.

4. Pathophysiology

Why obstruction destroys the uterus

1. Contractions continue against an immovable block. The **upper segment** keeps contracting and **retracting** (its fibres get shorter and thicker with each contraction), while the passive **lower segment** is stretched thinner and thinner.
2. The junction between the thick upper segment and the thin lower segment rises and becomes visible or palpable as a transverse groove: the **pathological retraction ring (Bandl's ring)**. It can rise towards the umbilicus. **Bandl's ring means imminent rupture.**
3. Eventually the thinned lower segment tears. This is **uterine rupture**: fetal death, massive haemorrhage into the peritoneum, and maternal shock.
4. In multiparas, the uterus may instead go into **exhaustion (secondary inertia)**, so contractions stop. This is not "improvement". In primigravidas the uterus rarely ruptures and tends to become **tonically contracted**.

Why obstruction causes fistula. The impacted head **compresses the bladder base, urethra, vagina and rectum against the pubic symphysis and sacrum** for hours. This causes **ischaemic pressure necrosis**. The dead tissue sloughs 3–10 days later and leaves a **vesicovaginal (VVF) or rectovaginal (RVF) fistula**. Haematuria and an oedematous anterior lip of the cervix during labour are early warnings. The same ischaemia damages the pelvic nerves (**foot drop**, from lumbosacral plexus or common peroneal nerve injury), the pubic bones and the vaginal walls (which later scar and cause **vaginal stenosis**). Together these make up the "obstructed labour injury complex".

Why obstruction kills the baby. Strong contractions without progress cause repeated reductions in placental blood flow, which leads to **hypoxia, acidosis and fetal distress**, and then **intrapartum stillbirth**. Prolonged rupture of membranes adds **ascending infection** (chorioamnionitis, neonatal sepsis).

Why the mother becomes septic, dehydrated and ketotic. Long labour without food or fluids leads to **dehydration and ketoacidosis** (a dry tongue, tachycardia, ketonuria, oliguria). Repeated vaginal examinations and long-ruptured membranes lead to **ascending infection** (fever, foul liquor). A tachycardic, febrile, dehydrated woman is a poor surgical candidate. That is why you **resuscitate before and during** the move to theatre.

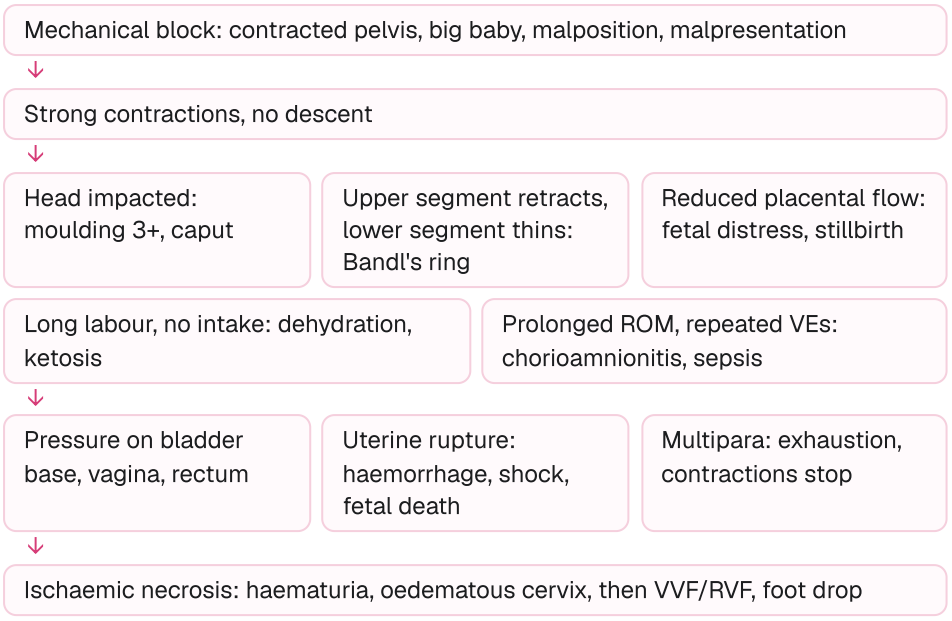
Why malposition and malpresentation obstruct: the diameters

Attitude of the head (term fetus)	Presenting diameter	Size	Result
Well-flexed vertex (OA)	Suboccipitobregmatic	**9.5 cm**	Fits easily
Deflexed vertex (OP)	Occipitofrontal	**11.5 cm**	Slow, may obstruct
Brow	Mentovertical	**13.5 cm**	Will **not** fit
Face (fully extended)	Submentobregmatic	**9.5 cm**	Fits **if mentoanterior**

Average gynaecoid pelvic inlet: transverse ~13 cm, AP ~11 cm.

- **OP position**: the head tends to be **deflexed**, so it presents the larger occipitofrontal diameter. The occiput has to rotate through 135 degrees (the "long arc") instead of 45. Labour is long, backache is severe, and the woman gets an urge to push early.
- **Deep transverse arrest**: the head descends in OT but fails to rotate at the level of the spines, often because of an android (heart-shaped) pelvis with prominent spines or poor contractions.
- **Face**: the neck is fully extended. **Mentoanterior** face can deliver because the chin rotates under the symphysis and the head is then born by **flexion**. In **mentoposterior** face the neck is already maximally extended and cannot extend further to negotiate the sacrum, so it is **undeliverable vaginally at term**.
- **Brow**: halfway between flexion and extension, it presents the **largest diameter (13.5 cm)**. At term it obstructs unless it flexes to a vertex or extends to a face.

- **Shoulder presentation**: no fetal part can engage, so it is **always obstructed** if left alone. This is **"neglected shoulder presentation"**, often with a **prolapsed arm**, and it causes rupture, especially in grand multiparas.
- **Breech**: the body (the smaller part) is born first. The **after-coming head** has **no time to mould** and can become trapped, especially in a preterm baby whose body can pass through an incompletely dilated cervix.



Feature links

- Mechanical block + strong contractions → no descent → head 3/5 or more palpable, prolonged labour crossing the action line.
- Skull compression in the pelvis → moulding 3+ (non-reducible overlap) and caput → a falsely low "station".
- Cervix compressed between head and pubis → oedematous anterior lip "hanging like a curtain", loosely applied.
- Retraction of the upper segment and thinning of the lower segment → Bandl's ring, tender stretched lower segment → rupture.
- Bladder pulled up and compressed → distended bladder above the symphysis, haematuria → later VVF.
- Rupture → contractions stop, presenting part recedes, fetal parts under the skin, shock, absent FHR.
- Repeated hypoxia → fetal tachycardia, bradycardia, meconium, death.
- Starvation and effort → dry tongue, sunken eyes, ketotic breath, ketonuria, oliguria.
- Ascending infection → fever, tachycardia, hot dry vagina, foul liquor.
- Nerve compression → foot drop after delivery.
- Deflexed OP head → anterior fontanelle easily felt anteriorly, backache, early urge to push.
- After-coming head with no time to mould → trapped head in breech (especially preterm).
- Fully extended neck in mentoposterior face → cannot extend further → obstruction.

5. Classification / Staging

Grading moulding and caput (partograph and LCG)

Grade	Moulding (skull bones at the sutures)	What it means
0	Bones separated, sutures felt easily	Normal
1+	Bones **just touching**	Normal adaptation
2+	Bones **overlapping but reducible** with gentle pressure	Watch closely

Grade	Moulding (skull bones at the sutures)	What it means
3+	Bones **overlapping and NOT reducible**	**Disproportion or obstruction**

Caput (scalp oedema) is graded 1+ to 3+ in the same way. Big caput **plus** 3+ moulding **plus** no descent means CPD until proven otherwise. Remember that caput can make the head *feel* lower than it is, so **always pair the vaginal station with the abdominal fifths palpable**.

Adequacy of contractions: **≥3 in 10 min lasting >40 s = adequate**; fewer and shorter = inefficient uterine action.

Engagement: fifths of the head palpable above the brim (Crichton method); **3/5 or more in the second stage = not engaged**.

Breech types

- **Frank (extended)**: hips flexed, knees extended. The commonest type, especially in primigravidas.
- **Complete (flexed)**: hips and knees flexed.
- **Footling or incomplete**: one or both hips extended, with a foot below the buttocks. Highest risk of cord prolapse.

Denominators: vertex = occiput; face = mentum (MA vs MP); breech = sacrum; shoulder = acromion.

6. Clinical Features

History

Take it while the cannula goes in. Each item has a reason.

Ask	Why it changes the plan
Age, parity, height	Adolescent or primigravida under ~150 cm: contracted pelvis likely; uterus tends to be tonic. **Multipara**: obstruction means **malpresentation or a big baby** (she has proven her pelvis), and she is at higher rupture risk
Time labour started; time membranes ruptured; colour of liquor	Hours of bladder compression predict **fistula**; ROM >18 hours plus fever means chorioamnionitis; meconium is a fetal warning
Where she laboured (home, TBA, HC II/III); who examined her; any injections or herbs ("medicine to push")	Herbal oxytocics or unsupervised oxytocin plus obstruction is the classic Ugandan route to **rupture**
Pushing for how long?	A long second stage with no delivery points to obstruction, DTA or a persistent OP
Sudden severe pain, then contractions stopped; feeling faint	**Uterine rupture** until proven otherwise
Fetal movements	Absent movements suggest IUFD, which changes the mode of delivery (CS vs destructive delivery)
Passing urine? Blood in it?	**Haematuria** = bladder compression (fistula warning) or rupture involving the bladder; anuria = dehydration, AKI or rupture
Previous CS or uterine surgery	A scarred uterus plus obstruction: **highest rupture risk**, go to theatre
Previous births: biggest baby, prolonged labours, instrumental births, stillbirths	Predicts CPD; the biggest previous baby "tests" the pelvis
ANC card: scan (lie, placenta), Hb, HIV, syphilis, blood group	Praevia (no VE until excluded), known breech, anaemia (tolerates bleeding poorly), PMTCT plan, anti-D

Ask	Why it changes the plan
Fever, rigors, foul liquor	Chorioamnionitis, sepsis: broaden antibiotics, prepare for a sick baby

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination (inspection, fundal height, lie, presentation, engagement in fifths, fetal heart), then the vaginal examination.

Introduction, consent and preparation

- Introduce yourself, confirm identity, explain, and gain consent; chaperone for VE.
- **Hand hygiene**; sterile gloves for VE; **one examiner** (repeated VEs cause infection).
- **Position and exposure**: left lateral between examinations; semi-recumbent for abdominal palpation, between contractions.

General inspection (look for exhaustion and sepsis)

Sign	Mechanism	What it means
Anxious, restless, exhausted; bearing down with no progress; or ominously quiet after hours of pain	Prolonged effort; rupture	Obstruction; rupture if suddenly quiet
Dry tongue, sunken eyes	Dehydration	Resuscitate
Ketotic breath	Starvation ketosis	Fluids, dextrose
Pallor	Anaemia, haemorrhage	Rupture, low reserve
Oedema	Pre-eclampsia can coexist	Check BP and urine
Blood or meconium on the pad	Bleeding; fetal compromise	Rupture/abruption; fetal distress

Vital signs: pulse, BP, RR, temperature, SpO_2 if available, **shock index** (HR ÷ SBP; ≥0.9 abnormal, ≥1.4 urgent). **Tachycardia (over 100/min), fever**, hypotension. A pulse over 100–110 with fever means sepsis or dehydration until proven otherwise; tachycardia with hypotension after a long labour means **rupture or haemorrhage** until proven otherwise.

Hands, face and neck

Cool peripheries and slow capillary refill (shock); conjunctival pallor (anaemia); dry mucous membranes (dehydration).

Obstetric abdominal examination (between contractions)

Step	Look / feel for	Document as
Inspection	Shape (uterus **wider than long** = transverse lie), scars, a groove running obliquely across the abdomen (**Bandl's ring**), distended bladder; flattened lower abdomen (OP)	"Bandl's ring 2 fingerbreadths below the umbilicus"
Fundal height	Size for dates; macrosomia, polyhydramnios	cm
Lie and presentation	Longitudinal cephalic / breech; transverse or oblique lie (head in a flank, empty pelvis); fetal limbs felt anteriorly (OP); deep groove between head and back (face)	e.g. "Longitudinal lie, cephalic"
Engagement: fifths of the head palpable above the brim (Crichton method)	**3/5 or more in the second stage = not engaged**, even if the vaginal station seems low	e.g. "3/5 palpable"

Step	Look / feel for	Document as
Lower segment	Tender? Stretched?	
Bladder	Distended suprapubic swelling, often pushed up above the symphysis	
Contractions (hand on the fundus for 10 min)	Number in 10 min and duration: **≥3 in 10 min lasting >40 s = adequate**; **tonic** uterus that never relaxes = danger	"4 in 10 min, 50 s"
Auscultation of the fetal heart (Pinard or Doppler, full minute after a contraction)	Absent, <110 or >160 bpm; heard in the flank (OP)	FHR
Signs of rupture	Fetal parts under the skin, uterus felt as a separate mass, peritonism, absent FHR, shock	

Vaginal examination

Only after placenta praevia is excluded (ANC scan or history).

Feel for	Obstruction pattern
Vulva, vagina	**Oedematous vulva**; **hot, dry vagina**; foul liquor or meconium
Cervix	Dilatation; **oedematous anterior lip, loosely applied** to the head ("hanging like a curtain")
Caput	Grade 0 to 3+ (size of the scalp swelling)
Moulding	0 separated, 1+ touching, 2+ overlapping but reducible, **3+ overlapping and not reducible**
Station	cm above or below the spines, always **paired with the fifths palpable** (caput makes the head feel lower than it is)
Position (sutures, fontanelles, denominator)	See the table below
Cord or limb	Prolapsed cord, prolapsed arm (**never pull**)
Pelvis (clinical pelvimetry)	Sacral promontory reachable, prominent ischial spines, subpubic arch narrower than 2 fingers, intertuberous diameter under 4 knuckles

How to find the position on VE

Landmark	What you feel	What it tells you
Sagittal suture	The long suture between the parietal bones	Its direction: transverse = OT; oblique = OA/OP; AP = direct OA/OP
Posterior fontanelle (lambda)	Small **triangle**, **3 sutures** (Y shape)	Marks the **occiput** (the denominator). Felt anteriorly = OA
Anterior fontanelle (bregma)	Large **diamond**, **4 sutures**	Easily felt = **deflexed head**; felt anteriorly = **OP**; felt centrally with orbital ridges = **brow**
Brow	Anterior fontanelle, supraorbital ridges, root of the nose	Brow presentation
Face	Mouth with hard gums, nose, malar bones (a **triangle**); the mouth sucks your finger	Denominator **mentum**: MA vs MP
Breech	Anus and ischial tuberosities in a **straight line**, sacrum, meconium on the glove	Denominator **sacrum**; feet felt = footling
Shoulder	Ribs, scapula, axilla (its closed apex points to the side of the fetal head), a hand	Denominator **acromion**; transverse lie

EXAM FAVOURITE

Hand versus foot on VE: a hand has fingers of unequal length and a thumb that can be abducted, and there is no heel. A foot has a heel, toes in a straight line and a big toe that does not abduct. **Mouth versus anus**: the mouth has hard gums and sucks your finger. The anus grips, and meconium stains the glove.

Completing the examination

- **Catheterise** and look at the urine: **blood-stained urine** in the bag; volume.
- Thank her, hand hygiene, plot findings on the partograph/LCG, and summarise.

EXAM FAVOURITE

The examiner will ask you to "present the VE". Always give all seven: **dilatation, cervical state (oedema, application), station, position, caput, moulding, liquor**, and then the **fifths palpable**. A student who says "8 cm" and stops has missed the diagnosis.

Red flags

RED FLAG

- **Bandl's ring**, a tender lower segment, or haematuria: rupture is imminent. Go to theatre now.
- **Sudden cessation of contractions**, shock, fetal parts under the skin, or loss of the presenting part on VE: **uterine rupture**. Resuscitate and do an emergency laparotomy.
- **Maternal fever over 38 °C**, tachycardia, or foul liquor: sepsis. Start IV antibiotics immediately.
- **Prolapsed cord**: category 1 emergency.
- **Prolapsed arm with transverse lie**: neglected shoulder presentation. **Never pull the arm.**
- **A scarred uterus** plus prolonged labour.
- **Fetal bradycardia**, or loss of the fetal heart.

RED FLAG

The clock starts at diagnosis: the Mulago standard is **delivery within 2 hours**. The history, examination and tests are done **while** the team prepares theatre, not before.

Thinking like a clinician

1. **Slow labour: is it the powers or an obstruction?** Look at the contractions. **Weak or infrequent contractions and no signs of obstruction** mean **inefficient uterine action**, which (in a primigravida with a cephalic presentation and no CPD) may be augmented with oxytocin. **Strong contractions and no descent**, with moulding 3+, caput and an oedematous cervix, mean **obstruction**, and oxytocin would be lethal.
2. **Never trust station alone.** A head at "0 station" with 3/5 palpable abdominally is **not engaged**. That is caput and moulding lying to you.
3. **In a multipara, "obstructed labour" is a malpresentation or a big baby until proven otherwise.** A multipara has already proven her pelvis with previous babies.
4. **Contractions stopping in a long labour is ominous**: think exhaustion or rupture, not "labour settling".
5. **The dead baby does not remove the emergency.** The mother still has an obstructed uterus that can rupture and become infected. A **CS for a dead baby** is sometimes the safest route when destructive delivery is not possible or safe. The decision depends on cervical dilatation, the station of the head, operator skill and signs of rupture.
6. **Think of fistula from the moment of diagnosis.** Catheterise at once, and document haematuria and the amount of urine. Keep the catheter in afterwards.

7. **Breech in labour at an HC III or IV**: check the gestation, the type of breech, the estimated fetal weight, how far labour has progressed and whether theatre is available. A multipara at 8 cm with a frank breech will often deliver before transfer, so **be ready to conduct an assisted breech**.
8. **Common student mistakes**:
    - reporting only the cervical dilatation without the **fifths palpable, moulding, caput and the state of the bladder**
    - suggesting oxytocin for "poor progress" without checking for obstruction
    - forgetting the catheter
    - pulling on a breech
    - calling a face a breech

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Obstructed labour	Strong contractions, no descent, moulding 3+, caput, Bandl's ring, oedematous cervix, haematuria	Clinical: partograph or LCG crosses the action line; head 3/5 or more palpable in the second stage
Inefficient uterine action (hypotonic)	Weak or infrequent contractions (fewer than 3 in 10 min, each under 40 s), no moulding 3+, the mother is not distressed	Tocography by hand: count contractions over 10 min
Uterine rupture	Sudden pain, then contractions stop, shock, fetal parts palpable, loss of station, vaginal bleeding, haematuria	Clinical; point-of-care ultrasound shows a fetus or free fluid outside the uterus
Placental abruption	Pain, a "woody" hard uterus, dark bleeding, shock out of proportion to visible loss, fetal distress or death	Clinical; ultrasound (retroplacental clot, not always visible)
False labour	Irregular contractions, no cervical change	Serial VE 2–4 hours apart
Full bladder impeding descent	Palpable suprapubic mass; descent improves after voiding	Catheterise
Malposition (OP) with normal progress	Backache, slower but steady progress	Serial VE; plot on the LCG
Deep transverse arrest	Head OT **at or below the spines**, ≤1/5 palpable, no obstruction signs	VE + abdominal fifths

8. Clinical Workup

The aim is to answer four questions quickly: **Is she (or is she about to be) ruptured or septic? Is this obstruction or just slow labour? What is presenting, and in which position? Is the baby alive?** Do not delay delivery waiting for investigations; send them as you prepare for theatre.

Bedside

Test	What you want	Why / what it changes	Available at
Urine via the catheter: volume and colour	Volume drained, **blood-staining**, then hourly output (target **≥30 mL/h**)	Haematuria = fistula warning; oliguria = dehydration, sepsis or haemorrhage	All levels

Test	What you want	Why / what it changes	Available at
Urine dipstick	**Ketones**, blood, protein	Ketosis confirms dehydration/starvation; proteinuria + high BP = pre-eclampsia	All levels
HemoCue / Hb	Baseline Hb	Anaemic women tolerate rupture and PPH poorly; request more blood	HC III and above
Random blood sugar	Hypoglycaemia after a long starvation labour	Treat before theatre	All levels
Bedside clotting test	Clot in a plain glass tube by 7 minutes	No clot = coagulopathy (rupture, abruption, sepsis): escalate, fresh blood/FFP	All levels
HIV and syphilis RDT (if not known)	Status	PMTCT for the baby; universal precautions	All levels
Fetal heart by Doppler/Pinard (CTG rarely available)	Rate, decelerations	Fetal distress raises urgency; absent FHR changes the delivery route	All levels
Partograph / LCG	Alert or action line crossed; 3+ moulding and descent plotted	The diagnostic "investigation" for prolonged and obstructed labour; send it with the referral	All levels
Clinical pelvimetry (antenatally, or at the first VE)	Sacral promontory reached, prominent spines, narrow subpubic arch (fewer than 2 fingers), intertuberous diameter (fewer than 4 knuckles)	Suggests a contracted pelvis; predicts but does not prove CPD, so labour progress is still the real test	All levels
Point-of-care ultrasound (if at hand, **never delaying theatre**)	Lie, presentation, FHR, placenta, free fluid	Confirms IUFD, malpresentation, praevia, rupture	HC IV and above with an operator

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC / Hb	Every woman, with the cannula	Low Hb; high WBC (with fever = infection; labour alone can raise WBC)	Cross-match more units; supports sepsis treatment	HC IV and above
Group and cross-match (≥2 units)	Every woman	Blood group; Rh status	Blood ready for CS/rupture/PPH; anti-D if Rh-negative with an Rh-positive baby	Blood banks at HC IV are unreliable; RRH/NRH usually have blood
U&E, creatinine	Long labour, oliguria, sepsis, rupture	High urea/creatinine (AKI), low K^+ or Na^+	Fluid plan; **gentamicin dose adjustment or avoidance**	Mostly RRH/NRH

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Ultrasound (formal)	Doubt about lie, presentation, fetal life or placenta, when it will not delay delivery; antenatal breech (cause, praevia, anomalies, liquor)	Malpresentation, IUFD, praevia, free fluid, fetus outside the uterus	Confirms rupture or IUFD; excludes praevia before VE or ECV	Most HC IVs and hospitals

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Blood culture	Temperature ≥38 °C or rigors	Organism and sensitivity	Tailor antibiotics after delivery	RRH/NRH; often unavailable
High vaginal swab / placental swab	Foul liquor, chorioamnionitis	Organism	Guides antibiotics for mother and baby	Limited availability
Coagulation (PT/APTT, fibrinogen, platelets)	Rupture, abruption, sepsis, a failed bedside clotting test	Prolonged times, platelets <100, fibrinogen <2 g/L	Blood products; senior anaesthetist	RRH/NRH only
Serum lactate / blood gas	Septic shock	Lactate raised	Severity marker	Rarely available outside NRH ICUs
CTG	Live fetus, cord prolapse, before and after ECV	Abnormal trace	Urgency of delivery	Referral hospitals
Paired cord gases	After cord prolapse or asphyxia	Acidosis	Neonatal care, audit	NRH/private Verify *availability*

Putting it together

Model one-line summary: "A 19-year-old primigravida at term, referred after 30 hours of labour, **exhausted, febrile and tachycardic**, with the head **3/5 palpable**, a **Bandl's ring**, **moulding 3+**, an oedematous cervix at 8 cm in ROT, fetal tachycardia and **blood-stained urine**: obstructed labour with impending rupture, chorioamnionitis and fetal distress."

Problem list

1. Obstructed labour (cause: CPD with a deflexed ROT head)
2. Impending uterine rupture (Bandl's ring, tender lower segment)
3. Chorioamnionitis / sepsis
4. Dehydration and ketosis
5. Fetal distress (live fetus)
6. High fistula risk (haematuria)

Working diagnosis and plan: obstructed labour in a live fetus → **resuscitate, catheterise, antibiotics, cross-match, emergency CS within 2 hours**, catheter for 14 days.

Model case presentation (ward round)

> "Mrs N is a 19-year-old primigravida at 40 weeks, referred from an HC III after 30 hours of labour, with membranes ruptured 20 hours ago. She is exhausted, febrile at 38.2 °C, with a pulse of 118 and a dry tongue. The abdomen shows a cephalic presentation with 3/5 palpable, a Bandl's ring at 2 fingerbreadths below the umbilicus, and a tender lower segment. The FHR is 170 bpm. On VE the cervix is 8 cm and

> oedematous, the position is ROT, and there is caput 3+ and moulding 3+. The catheter drained 100 mL of blood-stained urine. **Summary: obstructed labour with impending uterine rupture, chorioamnionitis and fetal distress.** I have started Ringer's lactate, sent a cross-match, given IV ceftriaxone and metronidazole, and informed theatre for an emergency CS. The catheter will remain in for 14 days because of the fistula risk."

Worked example 1: confirming obstruction and naming the position

A 22-year-old P0+0 at term, 10 hours in the active phase, pushing for 2 hours.

- **Abdomen**: cephalic, **3/5 palpable**, strong contractions **4 in 10 min lasting 50 s**, bladder palpable above the symphysis, no Bandl's ring. FHR 150.
- **VE**: fully dilated, **station +1**, **caput 2+**, **moulding 3+** (the parietal bones overlap and cannot be reduced), **sagittal suture transverse**, a **diamond with 4 sutures on the mother's right**, the triangle on her left.

Interpretation:

- The posterior fontanelle (occiput) is on her left and the sagittal suture is transverse: **LOT**.
- The anterior fontanelle is easily felt: the head is **deflexed**.
- "Station +1" but **3/5 palpable** means the head is **not engaged**; the low station is **caput and moulding** lying to you.
- Strong contractions + no descent + moulding 3+ = **obstruction, not DTA**. DTA requires the head **at or below the spines with no obstruction** (≤1/5 palpable).
- **Plan**: no oxytocin, no ventouse; resuscitate, catheterise and do a **CS**. Warn the surgeon about an **impacted head** (a push-from-below assistant ready).

Worked example 2: the multipara with an arm in the vagina

A 34-year-old P6 arrives from home with an arm hanging from the vulva, labour for 18 hours. Pulse 124, BP 96/60 (shock index 1.3), temperature 38.4 °C.

- **Abdomen**: uterus **wider than long**, head in the left flank, no presenting part in the pelvis, **Bandl's ring** halfway to the umbilicus, **no fetal heart**.
- **VE**: cervix 9 cm, ribs and a **closing axilla** pointing to her left, the arm prolapsed, oedematous vulva.

Interpretation: **neglected shoulder presentation (transverse lie), IUFD, impending (or actual) rupture and sepsis** in a grand multipara. Shock index >0.9 = significant compromise: look hard for rupture (tender uterus, free fluid on POCUS). **Do not pull the arm.**

Plan: two cannulae, fluids, cross-match, catheter, **IV ampicillin + gentamicin + metronidazole** (or ceftriaxone + metronidazole per protocol), senior obstetrician. **CS (or laparotomy if ruptured)** is usually safest; decapitation only in expert hands with no rupture. Catheter 14 days; watch for PPH.

9. Management

Immediate / emergency

Obstructed labour: the emergency algorithm (the Mulago standards)

1. **Look at her from the door**, then **recognise and call for help**: the senior midwife, the doctor on call, the anaesthetist and theatre. **Stop any oxytocin infusion** (and ask about herbal "medicine to push").
2. **A/B**: **left lateral** position, to relieve aortocaval compression. Give **oxygen** if in distress or shocked.
3. **C**: **two large-bore cannulae (16–18 G)**. Take blood for **Hb, group and cross-match (at least 2 units)**, urea and electrolytes, and blood culture if febrile. **Rehydrate**: **Ringer's lactate or 0.9% saline, 1 L fast**, then

continue to give **at least 1.5 L** before and during surgery. Titrate to pulse, BP and urine output (target over 30 mL/hour). If she is shocked, give boluses and treat as haemorrhage or sepsis. Calculate the shock index.

4. **Catheterise**: **Foley 14–16 F**. Empty the bladder and **note the colour** (haematuria) and volume. Leave it on free drainage. An empty bladder is safer at CS (less bladder injury) and it protects the bladder base.
5. **D/E**: conscious level, glucose, temperature. **Antibiotics**: IV broad-spectrum antibiotics now (drug table).
6. **Assess**: is the fetus alive? What is the cervical dilatation? How many fifths are palpable? Are there signs of rupture? What is the presentation?
7. **Decide the mode of delivery and aim to deliver within 2 hours.**
8. **At an HC III**: do all of the above, write a referral note with the partograph or LCG, and **refer immediately with a midwife escort**. Do not wait for a "trial" of anything.

> **DRUG DOSING**
> **Oxytocin is contraindicated in obstructed labour.** Augmenting contractions against a mechanical block ruptures the uterus, especially in multiparas and in scarred uteri.

Cord prolapse: algorithm (RCOG GTG 50)

1. **Call for help**, note the time, and **stop any oxytocin**.
2. **Do not handle the cord** (it causes vasospasm). If it is outside the vagina, replace it gently or cover it with warm, saline-soaked gauze.
3. **Relieve compression**: push the presenting part up with fingers in the vagina, or **knee–chest position**, or the **exaggerated Sims (left lateral, head down, pillow under the hip)** position, especially during transfer.
4. **Bladder filling** if a delay is expected (e.g. transfer from an HC III or IV): insert a Foley, **instil 500–750 mL of 0.9% saline**, and clamp it. **Empty the bladder before the uterus is opened.**
5. **Tocolysis**: **terbutaline 0.25 mg SC** if there is a delay and the CTG is abnormal.
6. **Deliver**:
   - **Category 1 CS** (aim for delivery **within 30 minutes**) if the fetus is alive and delivery is not imminent.
   - If the cervix is **fully dilated** and delivery is quick and safe: an **assisted vaginal birth** (ventouse or forceps) or a breech extraction by a skilled operator.
   - **If the fetus is dead**: vaginal delivery.
7. Have a neonatal resuscitation team ready. Take **paired cord gases** if possible. Debrief the family.

Prevention of cord prolapse: do not do an ARM when the head is high or the lie is unstable. Admit women with a transverse or unstable lie from 37 weeks.

Shoulder dystocia: HELPERR. Suspect it when the head delivers and then **retracts against the perineum ("turtle sign")**. Do not panic, **do not pull**, and **do not use fundal pressure**. Note the time the head was delivered.

Step	Action	Why it works
Help	Call the midwife, obstetrician, paediatrician and anaesthetist. Stop the mother pushing	You need hands, and pushing further impacts the shoulder
Evaluate for episiotomy	Makes room for internal manoeuvres. It does **not** free the shoulder on its own	The obstruction is bony, not soft tissue
Legs: **McRoberts**	Lay her flat and **hyperflex both thighs onto the abdomen**	Flattens the lumbosacral angle, rotates the symphysis upwards and increases the AP diameter. It works alone in up to about 90% of cases

Step	Action	Why it works
Pressure: **suprapubic**	An assistant pushes on the **posterior aspect of the anterior shoulder**, from the side of the fetal back, towards the fetal chest (continuous, then rocking)	Adducts and rotates the anterior shoulder into the wider oblique diameter
Enter (internal rotation)	**Rubin II** (push behind the anterior shoulder), **Woods' screw** (push on the front of the posterior shoulder), **reverse Woods**	Rotates the shoulders into the oblique diameter
Remove the posterior arm	Follow the posterior arm to the elbow, flex it, and sweep the hand across the chest and out	Reduces the shoulder width by 2–3 cm
Roll	**All-fours (Gaskin)** position, then retry the manoeuvres	Gravity and changes in pelvic diameters

Last resorts (rare): posterior axillary sling traction, deliberate clavicle fracture, **Zavanelli** (replacing the head followed by CS), symphysiotomy.

Afterwards: check the baby for **brachial plexus (Erb's) palsy**, fractured clavicle or humerus, and HIE; check the mother for **PPH** and **third- or fourth-degree tears**; document the **time of each manoeuvre, who did what, and which shoulder was anterior**.

Definitive

Delivery in obstructed labour

- **Live fetus**: **caesarean section**. Get the head out with care. A **deeply impacted head** may need an assistant to **push it up from below** (a "push method" with the hand in the vagina), or the surgeon can deliver the breech first (the reverse breech or "pull" method). Extend the incision to a J or inverted T if needed. Watch for lateral extension of the incision into the uterine vessels.
- **Dead fetus, cervix fully dilated, head 3/5 or less above the brim, no rupture, experienced operator**: consider **craniotomy** or another destructive delivery to avoid a scar in a woman who may labour again far from a theatre. Otherwise do a **CS**.
- **Signs of rupture**: **laparotomy** now, with repair or hysterectomy.
- **Neonatal team at delivery**: expect asphyxia and sepsis. Follow HBB/NRP.
- **After delivery**: active management of the third stage with **oxytocin 10 IU IM/IV**. Be vigilant for PPH, because the uterus is exhausted and atonic.

Destructive procedures and symphysiotomy (in context)

- **Craniotomy** (perforating the fetal skull to collapse it): for a **dead fetus** in cephalic presentation, with the **cervix fully dilated** (or at least 7 cm in expert hands), the **head 3/5 or less above the brim**, and **no signs of rupture**. **Decapitation** is for a neglected shoulder presentation with a dead fetus. Both need training. After a destructive delivery, **explore the uterus and genital tract** for rupture and tears, and **catheterise for at least 7–14 days**.
- **Symphysiotomy** (dividing the symphysis pubis fibres under local anaesthesia to enlarge the pelvis by 2–3 cm): historically used for a **live fetus, vertex, cervix fully dilated, head 3/5 or less above the brim, mild CPD, with no CS available**. It is **rarely done in Uganda today** because of long-term pelvic pain, walking difficulty, and urethral or bladder injury. It is examined as a concept, so know the indications and why it has fallen out of favour.

> **NOTE**
> The WHO and MSF texts still describe these procedures for settings with no safe CS. In a Ugandan hospital with theatre, the answer to "live baby, obstructed labour" is almost always **caesarean section**.

Malpositions

Occipito-posterior (OP)

1. **Diagnosis**: flattened lower abdomen, with fetal limbs felt anteriorly; FHR heard in the flank; backache; early urge to push; on VE, the **anterior fontanelle is easily felt anteriorly** and the posterior fontanelle posteriorly.
2. **First stage**: support, mobility, hydration, analgesia, an empty bladder. Most (about 90%) rotate to OA. If contractions are weak **and there are no signs of obstruction**, augmentation may be considered (in a primigravida, per protocol).
3. **Second stage**: allow time. Many deliver spontaneously as **"face to pubes"** (a direct OP), which needs a larger episiotomy. If the second stage is delayed with the head at or below the spines and no obstruction: **ventouse** (with rotation, e.g. a Kiwi OmniCup posterior cup) or **manual rotation** by a skilled operator. If the head is **high (more than 1/5 palpable) or there are signs of obstruction: CS.**

Deep transverse arrest

- The head is in OT at or below the spines and fails to rotate.
- **Check for obstruction first**: moulding 3+, caput, fifths palpable.
- **If there is no obstruction**: **rotational vacuum delivery**, or **manual rotation to OA** followed by vacuum; **Kielland's forceps** only in expert hands.
- **If there is obstruction or the attempt fails**: CS. Do not persist with a failed instrumental delivery.

Breech presentation

Antenatal plan (from 36 weeks)

1. **Confirm by ultrasound**. Look for the causes: placenta praevia, anomalies, fibroids, uterine anomaly, liquor volume.
2. **Offer ECV (external cephalic version)**: from **36+0 weeks in nulliparous** and **37+0 weeks in multiparous** women (RCOG GTG 20a); success is about **50%** overall (higher in multiparas); tocolysis with **terbutaline 0.25 mg SC** improves success; do a **CTG before and after**; give **anti-D** if she is Rh-negative; do it where emergency CS is available.
3. **Contraindications to ECV**: CS is indicated anyway (e.g. placenta praevia); APH within the last 7 days; abnormal CTG; a major uterine anomaly; ruptured membranes; multiple pregnancy (except the second twin). **Relative contraindications**: previous CS (discuss), FGR, pre-eclampsia, oligohydramnios, an unstable lie.
4. If ECV fails or is declined, counsel about **planned CS versus planned vaginal breech birth**. The **Term Breech Trial** (2000) found planned CS reduced perinatal mortality and serious neonatal morbidity, with no difference in maternal serious morbidity in the short term. However, a CS scar is dangerous in a future pregnancy in a low-resource setting (rupture, praevia, accreta).

Favourable criteria for vaginal breech birth: a frank or complete breech with a **flexed head**; an estimated fetal weight of about **2.0–3.8 kg**; an adequate pelvis; no other indication for CS; no footling presentation; a **skilled attendant** present; **spontaneous** labour that progresses well.

Assisted vaginal breech delivery: step by step

1. Full dilatation confirmed. Empty the bladder. Have the neonatal resuscitation set, a paediatrician or skilled midwife, and a warm towel ready. The mother is in the lithotomy position at the edge of the bed.
2. The mother pushes. **"Hands off the breech"** until the **umbilicus** is born. Traction causes **extension of the arms and head**. Consider an **episiotomy** as the buttocks distend the perineum.
3. Legs: if they are extended, flex the knee (**Pinard's manoeuvre**: press the popliteal fossa to flex the knee, then sweep the leg out).
4. Pull down a loop of the cord gently. Keep the **sacrum anterior** and hold the baby by the **bony pelvis** (thumbs on the sacrum, fingers on the iliac crests), never the abdomen (liver and kidney injury).

5. **Arms**: if they are not delivered spontaneously, use **Lovset's manoeuvre**. Rotate the baby 180 degrees, keeping the back anterior, to bring the posterior shoulder under the symphysis, and sweep that arm down across the face. Then rotate 180 degrees back the other way to deliver the other arm.
6. **After-coming head**: let the baby hang for a few seconds (**Burns–Marshall**) until the nape of the neck (hairline) is seen. Then either:
   - **Mauriceau–Smellie–Veit (MSV)**: the baby lies on your forearm, with your middle finger in the mouth or (preferably) index and ring fingers on the **malar bones** to **flex** the head. The other hand's middle finger pushes the occiput up and the other fingers pull on the shoulders. An assistant applies **suprapubic pressure** to keep the head flexed. Deliver slowly by flexion.
   - **Burns–Marshall**: grasp the feet and swing the body up in an arc over the mother's abdomen while the head delivers slowly. Guard the perineum.
   - **Forceps to the after-coming head** (Piper's, or ordinary obstetric forceps) if skilled.
7. **Delivering the head slowly** prevents a sudden decompression, which causes intracranial haemorrhage.
8. **Trapped head in a preterm baby** (cervix not fully dilated): **incise the cervix (Dührssen incisions at 2, 10 and 6 o'clock)**. Symphysiotomy is a last resort.

> **EXAM FAVOURITE**
> Viva: "Why hands off?" Answer: traction before the umbilicus is born makes the arms extend over the head (nuchal arms) and deflexes the head. The fetus is "delivered by the mother, not pulled by the doctor".

Face, brow, shoulder and compound presentations

Presentation	Diagnosis	Management
Face	A deep groove between the head and back abdominally. On VE: mouth, gums, nose and malar bones (a triangle); the mouth sucks your finger	**Mentoanterior**: allow labour. It usually delivers spontaneously by flexion. **No ventouse on the face.** Low forceps only if expert. **Mentoposterior** (persistent) or any obstruction: **CS**. Warn the parents that the face will be swollen and bruised
Brow	Head high, not engaged. On VE: anterior fontanelle, supraorbital ridges, root of the nose	**Persistent brow at term: CS.** Early in labour it may convert to face or vertex; watch briefly with low threshold. No oxytocin, no instruments
Shoulder / transverse lie	Uterus wider than long, no pole in the pelvis, head in one flank. On VE: ribs, axilla, scapula, or a **prolapsed arm**	**Antenatal**: look for the cause (praevia, fibroids, polyhydramnios). ECV from 37 weeks. Admit at about 37–38 weeks if the lie is unstable (cord-prolapse risk). **In labour with a live fetus: CS**. It may need a **vertical (classical or DeLee) uterine incision** if the lower segment is poorly formed or the back is down. **Neglected with a dead fetus**: CS is usually safest. Decapitation only by an expert. **Internal podalic version only for the second twin**
Compound	A hand or arm beside the head	Usually resolves. Do not push the arm up routinely; pinching the fingers may make the baby withdraw it. If obstructed or the cord prolapses: CS

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ringer's lactate / 0.9% saline	1 L fast, then titrate (at least 1.5 L peri-operatively)	IV	Continuous	Until resuscitated	Correct dehydration and ketosis. Monitor urine output (≥30 mL/h)

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ampicillin	2 g	IV	Every 6 hours	Until 48 h afebrile, then oral to complete 5–7 days	For prolonged rupture of membranes or amnionitis. Combine with gentamicin
Gentamicin	5 mg/kg	IV	Once every 24 hours	As above	Check renal function. Avoid if in AKI, or adjust the dose
Metronidazole	500 mg	IV	Every 8 hours	As above	Add for CS, rupture or foul liquor (anaerobic cover)
Ceftriaxone (common local alternative)	1–2 g	IV	Once daily	As above	Widely used with metronidazole in Ugandan hospitals. Follow your hospital protocol
Cefazolin or ampicillin (CS prophylaxis if no infection)	2 g	IV	Single dose, 15–60 min before skin incision	Once	Ampicillin is used at many Ugandan units
Oxytocin (third stage)	10 IU	IM or slow IV	Once after delivery	Then an infusion (20–40 IU in 1 L) if the uterus is atonic	**Contraindicated in obstructed labour** before delivery
Tranexamic acid (if PPH)	1 g	IV over 10 min	Repeat once after 30 min if bleeding continues		Give within 3 hours of birth
Terbutaline	0.25 mg	SC	Once		ECV tocolysis; cord prolapse with delay and abnormal CTG
Anti-D immunoglobulin	Per local vial Verify *dose not stated in source*	IM	Once		Rh-negative: after ECV; after delivery if baby Rh-positive
Paracetamol	1 g	Oral / IV	6-hourly	Post-operative	
Morphine	5–10 mg	IM (or per protocol)	4-hourly	Post-operative	
Enoxaparin (thromboprophylaxis after CS, if available)	40 mg	SC	Once daily	Per risk	Obstruction, sepsis and CS are all VTE risk factors
Oral step-down (e.g. amoxicillin + metronidazole, or cefixime + metronidazole)	Standard doses	Oral		To complete 5–7 days	After 48 hours afebrile
0.9% saline for bladder filling (cord prolapse)	500–750 mL	Via Foley	Once, clamp	Until theatre	Empty before the uterus is opened

UGANDA CONTEXT

In the Mulago audit, and in other Ugandan hospital protocols, a woman with obstructed labour gets **IV fluids, IV antibiotics, observations, a catheter and a cross-match, and is delivered within 2 hours**. In practice most Ugandan units use **ceftriaxone plus metronidazole** or **ampicillin plus gentamicin plus metronidazole**. Learn the regimen on your ward and follow the current **Uganda Clinical Guidelines 2023** and the **Essential Maternal and Newborn Clinical Care Guidelines**.

Referral criteria

Level	What to do	Refer up when
Home / TBA / HC II	Recognise prolonged labour and danger signs	Any labour not progressing → HC III/IV
HC III (no theatre)	Rehydrate, catheterise, first dose of antibiotics, partograph/LCG; assisted breech if delivery is imminent; cord prolapse first aid (bladder filling, knee–chest or exaggerated Sims, terbutaline)	**Immediately** for prolonged or obstructed labour, malpresentation not delivering, cord prolapse → HC IV/hospital, with a midwife escort and the partograph/LCG. **Never "a little oxytocin"**
HC IV / General Hospital	Resuscitate and do the CS on site; laparotomy for rupture; assisted breech, ECV where CS is available	Needs ICU, blood components, fistula repair, specialist surgery → RRH/NRH
RRH / NRH (Mengo, Mulago)	Blood components, ICU, fistula team, specialist care	n/a

Monitoring and follow-up

- Pulse, BP, temperature, urine output hourly during resuscitation; target urine **≥30 mL/h**.
- Continue IV antibiotics until **48 hours afebrile**, then oral to complete **5–7 days**.
- Watch for **PPH** (exhausted, atonic uterus), **sepsis** and **ileus**.

After delivery: preventing fistula and complications

1. **Leave the catheter in for 7–14 days** after prolonged or obstructed labour, or if there was haematuria. Many fistula units prefer 14 days. Encourage oral fluids of at least 3–4 L a day so the urine stays dilute and flushes the bladder.
2. If urine leaks per vaginam despite the catheter, **examine with a speculum and do a dye test**. A small fresh fistula may close with **continuous catheter drainage for 4–6 weeks**. Refer to the fistula team (see the VVF/RVF topic).
3. Monitor for **foot drop** (common peroneal or lumbosacral nerve injury): refer to physiotherapy.
4. **Counselling**: explain what happened. If the pelvis is contracted, the next delivery should be an **elective CS in hospital** at 39 weeks. **Birth spacing and family planning** matter.
5. **Mourning** if the baby died. Offer psychosocial support.

10. Complications

Disease-related

Maternal	Fetal / neonatal
Uterine rupture, haemorrhagic shock, death	Intrapartum stillbirth

Maternal	Fetal / neonatal
PPH (atony after a prolonged labour, tears)	Birth asphyxia, HIE, cerebral palsy
Puerperal sepsis, peritonitis, septic pelvic thrombophlebitis	Neonatal sepsis (prolonged rupture of membranes)
Obstetric fistula (VVF, RVF), vaginal stenosis	Intracranial haemorrhage (breech), cephalhaematoma
Foot drop (lumbosacral or common peroneal nerve injury)	Brachial plexus injury (shoulder dystocia, breech), clavicle or humerus fracture
AKI, VTE	Cord prolapse causing hypoxia
Secondary infertility, psychological trauma, stigma	Facial oedema and bruising (face presentation)

Treatment-related

Complication	Cause
Uterine rupture	Oxytocin or herbal uterotonics given in obstruction
Extension of the CS incision into the uterine vessels, PPH	Delivering an impacted head
Bladder injury at CS, ileus	Surgery in obstructed labour (hence an empty bladder)
Uterine and genital tract tears	Destructive delivery (explore afterwards)
Long-term pelvic pain, walking difficulty, urethral or bladder injury	Symphysiotomy
Intracranial haemorrhage	Instruments; rapid delivery of the after-coming head
Liver and kidney injury	Holding a breech by the abdomen
Nuchal arms, deflexed head	Traction on a breech before the umbilicus
Transient FHR changes, abruption, emergency CS (about 1 in 200)	ECV
Scar risks in future pregnancy (rupture, praevia, accreta)	Caesarean section

11. Prognosis

- With early recognition, resuscitation and CS within 2 hours, mother and baby usually survive; the harm in Uganda comes from **delay** (the three delays) before arrival.
- Uterine rupture carries high maternal mortality and almost always fetal death.
- Fistula risk rises with hours of impaction; a small fresh fistula may close with 4–6 weeks of catheter drainage.
- Most OP positions (about 90%) rotate to OA; mentoanterior face delivers vaginally; brow and mentoposterior face at term do not.
- ECV succeeds in about 50%.

12. Prevention

- **ANC**: identify risk (adolescent, short stature, previous CS, macrosomia, malpresentation at 36 weeks); scan breech for the cause; offer ECV; birth preparedness and delivery in a facility with theatre for high-risk women.
- **Labour**: plot every labour on the **partograph or LCG** and act on the alert/action lines; refer early from HC III; never augment without excluding obstruction; avoid herbal uterotonics.
- **Cord prolapse**: no ARM with a high head or unstable lie; admit transverse/unstable lie from 37 weeks.

- **After obstructed labour**: catheter 7–14 days; elective CS next time if the pelvis is contracted; family planning and birth spacing.
- **Community**: address the three delays; delay first pregnancy beyond adolescence.

13. Special Populations

- **Adolescents and short women (<150 cm)**: contracted pelvis likely; deliver in a facility with theatre.
- **Grand multiparas**: malpresentation, transverse lie and rupture risk; uterus prone to exhaustion and rupture rather than tonic contraction.
- **Scarred uterus**: obstruction → highest rupture risk; go to theatre.
- **Preterm breech**: the body can pass an incompletely dilated cervix and trap the head (Dührssen incisions).
- **Second twin**: internal podalic version is acceptable only here.
- **IUFD**: destructive delivery may be safer than a scar in a woman far from theatre, if criteria are met and the operator is skilled.
- **Rh-negative**: anti-D after ECV and after delivery if the baby is Rh-positive.
- **HIV**: PMTCT plan for the baby; universal precautions.

14. Key Points

High-yield points

- Strong contractions + no descent + moulding 3+ = obstruction; never oxytocin.
- Pair station with fifths palpable: 3/5 or more = not engaged.
- Mulago standards: fluids, antibiotics, observations, catheter, cross-match, delivery within 2 hours.
- Bandl's ring, tender lower segment, haematuria = imminent rupture.
- Catheter for 7–14 days after obstructed labour.
- Multipara obstructed = malpresentation or big baby.
- Brow and mentoposterior face at term = CS; mentoanterior face can deliver.
- Breech: ECV from 36/37 weeks; hands off until the umbilicus; Lovset, then MSV or Burns–Marshall.
- Cord prolapse: don't handle; elevate the presenting part; bladder 500–750 mL; category 1 CS.
- Shoulder dystocia: HELPERR; McRoberts + suprapubic pressure; never fundal pressure.

Numbers & doses

Item	Value
Delivery after diagnosis of obstructed labour (Mulago standard)	**Within 2 hours**
Resuscitation fluid	**Ringer's lactate or 0.9% saline 1 L fast**, then **at least 1.5 L** peri-operatively; urine **≥30 mL/h**
Catheter after prolonged/obstructed labour or haematuria	**7–14 days** (many fistula units: 14 days); oral fluids 3–4 L/day
Adequate contractions	**≥3 in 10 min**, each **>40 s**
Not engaged	**≥3/5 palpable** abdominally
Moulding 3+	Bones **overlap and cannot be reduced**

Item	Value
Presenting diameters	Well-flexed vertex **9.5 cm**; OP (occipitofrontal) **11.5 cm**; brow (mentovertical) **13.5 cm**; face (submentobregmatic) **9.5 cm**
Fontanelles	Anterior = **diamond, 4 sutures**; posterior = **triangle, 3 sutures**
Antibiotics (infected / prolonged ROM)	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly** (or ceftriaxone 1–2 g IV daily + metronidazole, per local protocol)
Oxytocin in the third stage	**10 IU IM/slow IV**; oxytocin **in labour is contraindicated**
ECV	From **36 weeks (nullip) / 37 weeks (multip)**; ~**50%** success; terbutaline **0.25 mg SC**
Cord prolapse bladder filling	**500–750 mL saline**; category 1 CS within **30 min**
Favourable EFW for vaginal breech	About **2.0–3.8 kg**
McRoberts alone resolves shoulder dystocia	Up to about **90%**
Shock index	**≥0.9 abnormal; ≥1.4 urgent**
Prolonged ROM	**>18 hours**
Dührssen incisions	**2, 10 and 6 o'clock**

Exam traps

- **"Poor progress, so start oxytocin."** Wrong until you have excluded obstruction: check contractions, fifths palpable, moulding and caput. Oxytocin + obstruction = rupture.
- **Trusting station alone.** A head at "0 station" with **3/5 palpable** is not engaged. Caput and moulding make it feel low.
- **Contractions stopping = labour improving.** No: exhaustion (secondary inertia) or **rupture**.
- **Dead baby = no emergency.** The mother can still rupture and become septic; she still needs resuscitation and delivery.
- **Ventouse on a face** or instruments on a brow. Never; mentoposterior face and persistent brow at term = **CS**.
- **Pulling a prolapsed arm** or pulling on a breech before the umbilicus. Never.
- **Calling it deep transverse arrest** when the head is 3/5 palpable. DTA is at or below the spines **without** obstruction; high OT with moulding 3+ is obstruction.
- **Fundal pressure in shoulder dystocia.** Never; McRoberts plus suprapubic pressure first.

Questions seniors ask

Viva questions

1. **What are the signs of obstructed labour? Maternal**: dehydration, tachycardia, fever, exhaustion, Bandl's ring, a tender lower segment, a distended bladder, haematuria, an oedematous vulva and cervix. **Fetal**: moulding 3+, caput, a high head, distress or death.
2. **Why is oxytocin contraindicated?** Contracting against a fixed obstruction thins the lower segment until it ruptures.
3. **Why does a mentoposterior face not deliver?** The neck is already maximally extended and cannot extend further around the sacrum. A mentoanterior face delivers by flexion.
4. **Which presenting diameter is largest?** Mentovertical (brow), 13.5 cm.
5. **Name three manoeuvres for the breech arms and head.** Lovset (arms), Mauriceau–Smellie–Veit and Burns–Marshall (head). Also Pinard (legs).
6. **When is ECV offered?** From 36 weeks in nulliparous women and 37 weeks in multiparous women. It is about 50% successful.

7. **Bladder volume for cord prolapse?** 500–750 mL of saline.
8. **First manoeuvres in shoulder dystocia?** McRoberts plus suprapubic pressure.

On the ward round

- **How do you tell obstruction from inefficient uterine action?** Obstruction: **strong** contractions, no descent, moulding 3+, caput, oedematous cervix, Bandl's ring, haematuria. Inefficient action: **weak** contractions (fewer than 3 in 10 min, <40 s) without those signs.
- **What are the six Mulago standards?** IV fluids, IV antibiotics, maternal vital-sign monitoring, bladder catheterisation, group and cross-match, delivery within 2 hours.
- **How do you know the head is deflexed?** The **anterior fontanelle** (diamond, 4 sutures) is easily felt; in a well-flexed head you feel only the posterior fontanelle.
- **Why catheterise before theatre and for 14 days after?** To empty the bladder (safer CS, less bladder injury), measure output and haematuria, and rest the ischaemic bladder base so a fistula may be prevented or heal.
- **When would you consider craniotomy instead of CS?** Dead fetus, cephalic, cervix fully dilated, head **3/5 or less** above the brim, no rupture, skilled operator.
- **What does a Bandl's ring mean?** The pathological retraction ring between the thick upper and the thinned lower segment: **imminent rupture**, go to theatre now.
- **A multipara is obstructed. What is the likely cause?** A malpresentation (shoulder, brow, face) or a big baby; she has proven her pelvis.
- **What are the first two manoeuvres in shoulder dystocia?** McRoberts and suprapubic pressure, after calling for help.

Mnemonics

- **Causes of obstruction: 3 Ps.** Passage (contracted pelvis, tumours), Passenger (big baby, malposition, malpresentation, anomaly, locked twins), Powers (usually strong in obstruction).
- **Signs of obstruction: "MOULD BH".** Moulding 3+, Oedematous cervix/vulva, Unengaged head (3/5+), Long labour, Distressed or dead fetus, Bandl's ring, Haematuria.
- **Shoulder dystocia: HELPERR.** Help, Evaluate for episiotomy, Legs (McRoberts), Pressure (suprapubic), Enter (rotational manoeuvres), Remove the posterior arm, Roll onto all fours.
- **Breech: "Hands off, back up, arms by Lovset, head by MSV."**
- **Face vs breech on VE:** the face makes a **triangle** (mouth and malar bones) and the breech makes a **line** (anus and ischial tuberosities).
- **"Chin to the back = CS."** A mentoposterior face will not deliver.

OSCE tips

- In a breech-mannequin station, **verbalise "hands off until the umbilicus"**, keep the **back anterior**, and **hold the pelvis, not the abdomen**.
- In a shoulder dystocia station, **say the time**, call for help, and avoid fundal pressure. Document everything.
- When you counsel for ECV, give the success rate (about 50%), the risks (small: transient FHR changes, abruption, need for emergency CS in about 1 in 200), the alternatives, and anti-D if Rh-negative.
- When you "present the VE", give all seven items plus the fifths palpable.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define obstructed labour.	Failure of descent of the presenting part **despite adequate uterine contractions**, because of a mechanical block (disproportion between the fetus and the birth canal).
2	List the clinical signs of obstructed labour (maternal and fetal).	**Maternal**: exhaustion, dehydration, tachycardia, fever, **Bandl's ring**, tender lower segment, distended bladder, **haematuria**, oedematous vulva and cervix. **Fetal**: head 3/5 or more palpable, **moulding 3+**, big caput, fetal distress or death.
3	Grade moulding 1+, 2+ and 3+.	1+ = bones touching. 2+ = overlapping but **reducible**. 3+ = overlapping and **not reducible** (suggests disproportion).
4	What is Bandl's ring and why does it matter?	A **pathological retraction ring**: the rising junction between the thick, retracted upper segment and the over-stretched thin lower segment. It signals **imminent uterine rupture**, so go to theatre now.
5	The six standards of care for obstructed labour (Mulago audit)?	1. IV fluids 2. IV antibiotics 3. Monitoring of maternal vital signs 4. Bladder catheterisation 5. Blood grouping and cross-matching 6. Delivery within 2 hours of diagnosis
6	Why is oxytocin contraindicated in obstructed labour?	Stronger contractions against a fixed block further thin the lower segment, which leads to **uterine rupture**, especially in multiparas and scarred uteri.
7	How long should the bladder catheter stay after prolonged or obstructed labour?	**7–14 days** (many fistula units prefer 14 days), with plenty of oral fluids. The aim is to prevent a fresh VVF, or help one heal.
8	Presenting diameters: OA vertex, OP (deflexed) vertex, brow, face.	OA vertex: suboccipitobregmatic **9.5 cm**. OP: occipitofrontal **11.5 cm**. **Brow: mentovertical 13.5 cm** (undeliverable at term). Face: submentobregmatic **9.5 cm**.
9	Mentoanterior vs mentoposterior face: management?	**Mentoanterior**: can deliver vaginally (by flexion). No ventouse. **Mentoposterior** (persistent): cannot deliver at term, so **CS**.
10	Types of breech presentation?	**Frank** (extended: hips flexed, knees extended; the commonest type). **Complete** (flexed: hips and knees flexed). **Footling/incomplete** (a foot below the buttocks; highest risk of cord prolapse).
11	When is ECV offered (RCOG), and how successful is it?	From **36+0 weeks in nulliparous** and **37+0 weeks in multiparous** women. It succeeds in about **50%** (higher in multiparas). Terbutaline tocolysis improves success. Give anti-D if Rh-negative.
12	Absolute contraindications to ECV?	CS indicated anyway (e.g. praevia), APH within 7 days, abnormal CTG, major uterine anomaly, ruptured membranes, multiple pregnancy (except the second twin).
13	Lovset's manoeuvre: what for and how?	Delivering **extended arms** in breech. Hold the pelvis, keep the back anterior, rotate 180 degrees so the posterior shoulder comes anterior below the symphysis, and sweep that arm down. Then rotate 180 degrees back for the other arm.

#	Question	Answer
14	Mauriceau–Smellie–Veit manoeuvre?	Delivers the **after-coming head** of a breech by **flexion**. The baby lies on your forearm, with fingers on the malar bones (or in the mouth) to flex the head. The other hand pushes the occiput up and pulls on the shoulders. An assistant gives suprapubic pressure.
15	Cord prolapse: how do you relieve compression while awaiting CS?	Manually elevate the presenting part, use the knee–chest or exaggerated Sims position, **fill the bladder with 500–750 mL saline** and clamp (empty it before opening the uterus), and give **terbutaline 0.25 mg SC**. Don't handle the cord.
16	HELPERR?	**H**elp, **E**valuate for episiotomy, **L**egs (McRoberts), **P**ressure (suprapubic), **E**nter (Rubin II, Woods screw), **R**emove the posterior arm, **R**oll (all-fours).
17	How does McRoberts work?	Hyperflexing the thighs flattens the lumbosacral angle and rotates the symphysis cephalad, increasing the effective AP diameter of the pelvic outlet. It alone resolves most shoulder dystocias.
18	Criteria for craniotomy in obstructed labour?	**Dead fetus**, cephalic, cervix fully dilated (7 cm or more in expert hands), head **3/5 or less above the brim**, **no signs of rupture**, and an experienced operator. Explore the tract afterwards and catheterise.
19	Transverse lie in labour with a live fetus: management?	**Caesarean section.** It may need a vertical or classical uterine incision if the lower segment is poorly formed or the back is inferior. Never pull a prolapsed arm. Internal podalic version is only for the second twin.
20	Obstructed labour injury complex: components?	VVF and/or RVF, vaginal stenosis and scarring, foot drop (nerve injury), pubic bone damage, secondary amenorrhoea or infertility, and social isolation.
21	Obstructed labour: what do you start in the first 5 minutes (while theatre is prepared)?	Call for help and **stop oxytocin**; left lateral; **2 large-bore cannulae** with blood for Hb, group and cross-match, U&E (culture if febrile); **Ringer's lactate or saline 1 L fast** (then at least 1.5 L); **Foley catheter** (note colour and volume); **first dose of IV antibiotics**; fetal heart. Aim to deliver **within 2 hours**.
22	On VE, how do you tell the anterior from the posterior fontanelle, and what does an easily felt anterior fontanelle mean?	**Posterior (lambda)**: small **triangle**, **3 sutures** (Y); it marks the occiput. **Anterior (bregma)**: large **diamond**, **4 sutures**. An easily felt anterior fontanelle = **deflexed head** (often OP); anterior fontanelle anteriorly = OP.
23	VE: station +1, but the head is 3/5 palpable abdominally. What is going on?	The head is **not engaged**. The low station is **caput and moulding**. Trust the abdomen: 3/5 or more palpable in the second stage with moulding 3+ = **obstruction**, and instrumental delivery is contraindicated.
24	What seven things must you report from a VE in suspected obstruction?	**Dilatation; cervix (oedema, how well applied); station; position (sutures, fontanelles, denominator); caput grade; moulding grade; liquor (colour, smell)**, plus the **fifths palpable** abdominally and the urine colour in the catheter bag.
25	Deep transverse arrest vs obstructed labour in OT position: how do you tell them apart?	**DTA**: fully dilated, head OT **at or below the spines** (≤1/5 palpable), **no** signs of obstruction: rotational ventouse or manual rotation. **Obstruction**: head high (≥3/5), moulding 3+, big caput, oedematous cervix, Bandl's ring, haematuria: **CS**.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 17-year-old primigravida is referred from an HC III after 26 hours of labour. She is exhausted, pulse 124/min, temperature 38.4 °C. The fetal head is 3/5 palpable, the cervix is 7 cm and oedematous, with moulding 3+ and caput 3+. The catheter drains blood-stained urine. FHR 168 bpm. Which is the MOST appropriate plan?

- A Start oxytocin augmentation and reassess in 2 hours
- B Perform artificial rupture of membranes and allow 4 more hours
- C Resuscitate with IV fluids, give IV antibiotics, keep the catheter, cross-match and perform emergency CS
- D Apply a ventouse to expedite delivery
- E Refer to the national referral hospital for a specialist opinion

2. A para 6 has been in labour at home for 18 hours. She suddenly felt a tearing pain and the contractions stopped. She is pale with BP 80/50 and pulse 130. Fetal parts are easily palpable under the abdominal wall, and no fetal heart is heard. What is the MOST likely diagnosis?

- A Uterine rupture
- B Placental abruption
- C Uterine inertia from exhaustion
- D Placenta praevia
- E Amniotic fluid embolism

3. A primigravida at 40 weeks has been in active labour with the cervix at 6 cm for 4 hours. She has 1 contraction in 10 minutes lasting 20 seconds. The head is 2/5 palpable, the position is LOA, moulding is 1+ and there is no caput. Membranes ruptured spontaneously with clear liquor. FHR is normal. What is the BEST next step?

- A Emergency caesarean section for obstructed labour
- B Give IV antibiotics and deliver within 2 hours
- C Symphysiotomy
- D Start oxytocin augmentation with close monitoring
- E Discharge and review tomorrow

4. At 7 cm, VE in a multipara reveals the anterior fontanelle, the supraorbital ridges and the root of the nose. The head is 4/5 palpable despite good contractions for 3 hours. What is the MOST appropriate management?

- A Allow labour to continue, expecting spontaneous delivery
- B Caesarean section
- C Ventouse delivery at full dilatation
- D Manual rotation to occipito-anterior
- E Oxytocin augmentation

5. Shortly after spontaneous rupture of membranes in a woman at 38 weeks with a footling breech, the midwife sees a loop of pulsating cord at the introitus. The cervix is 4 cm. The theatre is ready in 20 minutes. Which action is MOST appropriate while awaiting CS?

- A Push the cord back into the uterus as high as possible
- B Apply gentle traction to the legs to expedite delivery
- C Ask the mother to push with the next contraction
- D Start an oxytocin infusion to speed up dilatation
- E Elevate the presenting part manually and place her in the knee–chest position, avoiding handling the cord

6. During delivery of a 4.2 kg baby, the head delivers and then retracts against the perineum. Gentle traction fails. Help has been called. What is the NEXT manoeuvre?

A Fundal pressure
B Zavanelli manoeuvre
C McRoberts manoeuvre with suprapubic pressure
D Deliberate fracture of the clavicle
E Strong lateral traction on the head

7. A 25-year-old nulliparous woman at 36+2 weeks has an uncomplicated singleton frank breech confirmed on ultrasound. The placenta is fundal, liquor is normal and the CTG is reactive. She wishes to avoid CS. What should you offer?

A Elective CS at 37 weeks
B External cephalic version now, in a facility with theatre access
C Wait for spontaneous version until 41 weeks
D Induction of labour at 37 weeks for vaginal breech birth
E Internal podalic version in labour

8. A multipara is delivering a frank breech at an HC IV. The buttocks and legs have delivered spontaneously and the umbilicus is visible. The arms are not seen. What is the BEST next step?

A Pull on the legs to deliver the trunk quickly
B Apply fundal pressure
C Rotate the baby to sacrum posterior
D Hold the bony pelvis with the back anterior and perform Lovset's manoeuvre
E Perform an emergency CS

9. A primigravida at full dilatation has been actively pushing for 2 hours. VE: head at +1 station, 0/5 palpable, sagittal suture transverse, moulding 1+, no caput. Contractions are good and the FHR is normal. What is the MOST appropriate management of this deep transverse arrest?

A Rotational vacuum delivery (or manual rotation then vacuum) by a skilled operator
B Immediate CS
C Symphysiotomy
D Oxytocin and continued pushing for 3 more hours
E Craniotomy

10. Which feature BEST distinguishes a face presentation from a breech on vaginal examination?

A The presence of meconium on the glove
B A soft presenting part
C Presence of an orifice
D A high presenting part
E The mouth and malar prominences form a triangle, whereas the anus and ischial tuberosities lie in a straight line

11. A woman delivered by CS after 30 hours of obstructed labour. The fetus was stillborn. On day 2 her catheter bag contains light haematuria. What is the MOST important measure to prevent an obstetric fistula?

A Remove the catheter on day 2 and encourage voiding
B Start oral ciprofloxacin
C Keep the indwelling catheter for 7–14 days with high oral fluid intake
D Perform immediate fistula repair
E Pack the vagina with gauze

12. A para 5 at 39 weeks presents in early labour. The uterus is broad, no fetal pole is felt in the pelvis and the head is in the right flank. Membranes are intact, the cervix is 3 cm and the FHR is normal. What is the BEST management?

A Artificial rupture of membranes to stabilise the lie
B Caesarean section
C Allow labour and deliver by breech extraction
D Oxytocin augmentation
E Internal podalic version and extraction

13. In an assisted breech delivery, the nape of the neck is visible. The attendant lets the baby hang, then grasps the ankles and swings the body in an arc over the mother's abdomen while the head delivers slowly. Which manoeuvre is this?

A Lovset
B Pinard
C Mauriceau–Smellie–Veit
D Burns–Marshall
E Woods screw

14. At an HC IV in rural Uganda, a cord prolapse is diagnosed at 5 cm. The theatre team will take 45 minutes to arrive. The FHR is 100 bpm. Besides manual elevation, which measure is MOST useful?

A Fill the bladder with 500–750 mL of normal saline via a Foley catheter and give terbutaline 0.25 mg SC
B Give oxytocin 10 IU IM
C Give IV magnesium sulphate 4 g
D Deliver by vacuum extraction
E Encourage maternal pushing

15. A student asks why OP position causes long, painful labour. Which statement is MOST accurate?

A The fetal head is always flexed, presenting the smallest diameter
B The occiput must rotate only 45 degrees to reach the anterior position
C The head is often deflexed, presenting the occipitofrontal diameter (11.5 cm), and must rotate 135 degrees (the long arc)
D OP position is always associated with a contracted pelvis
E OP position prevents cervical dilatation completely

Answers & explanations

> **1. C** · Resuscitate with IV fluids, give IV antibiotics, keep the catheter, cross-match and perform emergency CS
>
> This is **obstructed labour** (moulding 3+, caput, a high head, an oedematous cervix, haematuria) with chorioamnionitis and fetal distress. Management is resuscitation, catheter, antibiotics, cross-match and **CS within 2 hours**. **A** Oxytocin against obstruction risks rupture. **B** She has been in labour for 26 hours; waiting worsens rupture and fistula risk. **D** Ventouse needs full dilatation and an engaged head, so it is contraindicated here. **E** A further referral adds delay; an HC IV or hospital with theatre should operate.

> **2. A** · Uterine rupture
>
> Sudden pain, then **cessation of contractions**, shock and **fetal parts under the skin** in a grand multipara after long labour is classic **uterine rupture**. **B** Abruption gives a hard, woody, tender uterus with the fetus inside, not easily palpable parts. **C** Inertia does not cause shock or palpable fetal parts. **D** Praevia gives painless bleeding, usually before labour. **E** Amniotic fluid embolism presents with sudden dyspnoea, collapse and DIC, without palpable fetal parts.

3. D · Start oxytocin augmentation with close monitoring

This is **inefficient uterine action** (weak, infrequent contractions) in a primigravida, **without signs of obstruction** (moulding 1+, no caput, the head descending). Oxytocin augmentation with fetal monitoring is appropriate. **A** There is no obstruction, and CS is premature. **B** Antibiotics are not indicated for clear liquor with a short rupture duration and no infection. **C** Symphysiotomy is for mild CPD with obstruction when CS is unavailable. **E** She is in active labour and needs care now.

4. B · Caesarean section

This is a **brow presentation** (mentovertical 13.5 cm) that persists with good contractions and no descent, so it will not deliver vaginally at term. **CS**. **A** A persistent brow obstructs. **C** Instruments are contraindicated on a brow. **D** Manual rotation is for OP or OT malposition, not brow. **E** Oxytocin in an undeliverable presentation risks rupture.

5. E · Elevate the presenting part manually and place her in the knee–chest position, avoiding handling the cord

Relieve cord compression by **elevating the presenting part**, using the **knee–chest** or exaggerated Sims position, filling the bladder if there is a delay, and giving terbutaline if needed, then a category 1 CS. **A** Handling or manipulating the cord causes vasospasm; only gently replace a cord that is outside the vagina. **B** Breech extraction at 4 cm risks head entrapment. **C** Pushing is useless at 4 cm and compresses the cord. **D** Oxytocin increases contractions and cord compression.

6. C · McRoberts manoeuvre with suprapubic pressure

In shoulder dystocia, after calling for help the first-line measures are **McRoberts plus suprapubic pressure**, which resolve most cases. **A** Fundal pressure worsens impaction and risks rupture and brachial plexus injury. **B** and **D** are last-resort salvage manoeuvres. **E** Excessive or lateral traction causes Erb's palsy.

7. B · External cephalic version now, in a facility with theatre access

RCOG recommends offering **ECV from 36+0 weeks in nulliparous women**. It is about 50% successful and reduces CS. **A** An elective CS without offering ECV denies her the option. **C** Spontaneous version becomes unlikely after 36 weeks in nulliparas, and waiting reduces ECV success. **D** Induction for breech is not first line. **E** Internal podalic version is only for a second twin.

8. D · Hold the bony pelvis with the back anterior and perform Lovset's manoeuvre

Extended arms are delivered by **Lovset's manoeuvre**: hold the **pelvis** (not the abdomen), keep the back anterior, and rotate so the posterior shoulder comes anterior below the symphysis. **A** Traction causes further arm extension and head deflexion. **B** Fundal pressure is not the technique and may extend the arms. **C** Sacrum posterior risks the chin catching on the symphysis. **E** The breech is half delivered, and CS is no longer possible or appropriate.

9. A · Rotational vacuum delivery (or manual rotation then vacuum) by a skilled operator

There are **no features of obstruction** (the head is engaged, moulding 1+, no caput). In deep transverse arrest, **rotational vacuum or manual rotation** by a skilled operator is appropriate. **B** CS is reserved for obstruction or a failed instrumental attempt. **C** Symphysiotomy is obsolete where CS is available and not indicated here. **D** A prolonged second stage risks fistula and fetal hypoxia. **E** Craniotomy is only for a dead fetus.

10. E · The mouth and malar prominences form a triangle, whereas the anus and ischial tuberosities lie in a straight line

The **triangle vs straight line** rule is the classic discriminator. **A** Meconium can be present in both (though it is fresher in breech). **B** Both are soft and oedematous. **C** Both have an orifice (mouth vs anus). **D** Height does not discriminate.

11. C · Keep the indwelling catheter for 7–14 days with high oral fluid intake

Continuous bladder drainage for **7–14 days** after prolonged obstructed labour lets the ischaemic bladder base heal. It prevents, or can close, a small fresh fistula. **A** Early removal loses this protection. **B** Antibiotics do not prevent pressure necrosis. **D** There is no fistula yet, and immediate repair of necrotic tissue fails. **E** Packing does not help and may add injury.

12. B · Caesarean section

A **transverse lie in labour** with a live fetus needs **CS**. (ECV could be tried only before labour or very early with intact membranes by experienced staff, and it is not among the options.) **A** ARM with a transverse lie risks **cord or arm prolapse**. **C** Breech extraction presupposes a breech. **D** Oxytocin leads to neglected shoulder and rupture. **E** Internal podalic version is reserved for a second twin because of the rupture risk.

13. D · Burns–Marshall

Burns–Marshall means hanging the baby, then swinging it up by the feet to deliver the after-coming head. **A** Lovset is for the arms. **B** Pinard is for extended legs. **C** MSV uses fingers on the malar bones to flex the head, with the baby on the forearm. **E** Woods screw is a shoulder dystocia manoeuvre.

14. A · Fill the bladder with 500–750 mL of normal saline via a Foley catheter and give terbutaline 0.25 mg SC

When delivery will be delayed, **bladder filling** (500–750 mL) keeps the presenting part off the cord, and **terbutaline** reduces contractions. Remember to empty the bladder at CS. **B** Oxytocin worsens compression. **C** $MgSO_4$ is not the recommended tocolytic here. **D** A vacuum is contraindicated before full dilatation. **E** Pushing at 5 cm is futile.

15. C · The head is often deflexed, presenting the occipitofrontal diameter (11.5 cm), and must rotate 135 degrees (the long arc)

The **deflexed OP head** presents the **occipitofrontal 11.5 cm** diameter and must travel the **long arc rotation of 135 degrees**, so labour is slow and backache is typical. **A** OP heads tend to be deflexed. **B** 45 degrees is the rotation from OA or oblique anterior positions. **D** OP is commonly associated with android or anthropoid pelvis shapes but is not always due to contraction. **E** The cervix dilates, just more slowly.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The adolescent referred after 30 hours

An 18-year-old primigravida is brought to Mengo labour suite on a boda boda from a village HC III, where she laboured for 24 hours after 6 hours at home with a TBA. She has had no ANC scan. She is groaning, febrile and says she has not passed urine since morning.

1. What focused history do you take, and why?

- **Duration of labour** and **time of rupture of membranes**: fistula and infection risk.
- **Any herbs or injections given** (oxytocics): rupture risk.
- **Fetal movements**.
- **Urine output and haematuria**.
- **Sudden pain or cessation of contractions** (rupture).
- **ANC**: HIV status, Hb, height, gestation.
- **Allergies**.
- **Previous uterine surgery** (none in a primigravida, but always ask).

2. Examination: T 38.6 °C, P 126, BP 100/60, dry tongue. Head 3/5 palpable, a groove 3 fingerbreadths below the umbilicus, a tender lower segment, FHR 172. VE: cervix 8 cm, oedematous anterior lip, caput 3+, moulding 3+, foul liquor. What is your diagnosis?

Obstructed labour with **Bandl's ring (impending rupture)**, **chorioamnionitis**, **dehydration** and **fetal distress**. Likely cause: CPD in an adolescent pelvis, possibly with a deflexed malposition.

3. List your immediate management in order.

1. Call the senior, anaesthetist and theatre. **No oxytocin.**
2. Left lateral position and oxygen.
3. Two large-bore IV lines; bloods for Hb, **group and cross-match 2 units**, U&E and culture.
4. **Ringer's lactate 1 L fast**, then titrate (at least 1.5 L).
5. **Foley catheter**: note haematuria and volume.
6. **IV antibiotics**: ampicillin 2 g plus gentamicin 5 mg/kg plus metronidazole 500 mg (or ceftriaxone plus metronidazole per protocol).
7. Paracetamol for fever.
8. **Emergency CS within 2 hours**, with a neonatal team present.

4. At CS the head is deeply impacted. How do you deliver it safely?

- Ask an assistant to **push the head up from below** (hand in the vagina) while you flex the head.
- Alternatively, deliver by the **breech first (reverse breech extraction)**.
- Extend the incision (J or inverted T) rather than tear the lower segment laterally.
- Anticipate **PPH** (atony, lateral extensions) and have oxytocin and tranexamic acid ready.

5. What postoperative orders prevent the major complications?

- **Catheter for 14 days** plus oral fluids of at least 3–4 L/day (fistula prevention).
- Continue **IV antibiotics until 48 hours afebrile**, then oral to complete 5–7 days.
- Monitor vitals and urine output.
- Watch for PPH, ileus, wound sepsis and **foot drop**.
- Thromboprophylaxis.
- Neonatal sepsis screen for the baby.
- Counsel on **elective CS in the next pregnancy** and on family planning.

Take-away: Recognise obstruction by its signs (moulding 3+, caput, Bandl's ring, haematuria), not by dilatation alone. Resuscitate, catheterise, give antibiotics and deliver within 2 hours. Keep the catheter to prevent fistula.

Case 2: Breech in advanced labour at an HC IV

A 28-year-old para 3 (three previous normal deliveries, largest baby 3.4 kg) arrives at an HC IV at 39 weeks, contracting strongly. The midwife finds a breech at 9 cm with membranes just ruptured and clear liquor. The medical officer is in theatre with another case, and referral would take 2 hours.

1. What do you assess to decide whether vaginal breech delivery is reasonable?

- **Type of breech** (frank or complete is favourable; footling is not).
- **Exclude cord prolapse** on VE.
- **EFW** (roughly 2.0–3.8 kg), and whether the head is flexed (clinically or on ultrasound).
- **Pelvic adequacy**: she has proven with 3.4 kg babies.
- **Progress**: rapid in a multipara.
- **FHR**.
- Whether **skilled help** and a **neonatal resuscitation** set are available.

At 9 cm in a multipara, transfer is unsafe: she may deliver on the road.

2. Describe your preparation.

- Call the most experienced person available.
- Prepare the **resuscitation corner** (bag-mask, warm towels).
- IV line and an empty bladder.
- Lithotomy position at the edge of the bed.
- Oxytocin 10 IU drawn up for the third stage.
- Episiotomy set and local anaesthetic.
- Forceps if available.
- Explain to the mother that she must push when asked.

3. The breech delivers to the umbilicus. The arms are extended. What do you do?

Hands off until the umbilicus, then gently loop the cord down. Hold the **bony pelvis** with thumbs on the sacrum and the back anterior. Perform **Lovset's manoeuvre**: rotate 180 degrees to bring the posterior arm anterior below the symphysis and sweep it down across the face, then rotate back 180 degrees for the other arm.

4. How do you deliver the after-coming head?

> Let the baby hang until the **nape of the neck (hairline)** is visible (**Burns–Marshall**). Then use **Mauriceau–Smellie–Veit**: the baby lies on your forearm, with fingers on the malar bones to flex the head, the other hand on the occiput and shoulders, and **an assistant giving suprapubic pressure**. Deliver **slowly** to avoid sudden decompression. Alternatively, swing the body up by the feet (Burns–Marshall).

5. The baby is floppy at birth. After resuscitation, what do you check in mother and baby?

> - **Baby**: Apgar, HBB (dry, stimulate, ventilate within the Golden Minute), injuries (humerus or clavicle fracture, brachial plexus palsy, abdominal organ injury, soft-tissue bruising), and hips (breech is a risk factor for DDH, so arrange hip ultrasound or examination).
> - **Mother**: active management of the third stage, tears, PPH.
> - **Documentation**: times and manoeuvres.

Take-away: Every clinician in Uganda must be able to conduct an assisted breech: hands off until the umbilicus, back anterior, hold the pelvis, Lovset for the arms, MSV or Burns–Marshall for a slow, flexed head.

Case 3: The head is out but the body won't come

A 34-year-old para 2 with gestational diabetes and an estimated fetal weight of 4.3 kg is delivering at Mengo. After a second stage of 90 minutes, the head delivers slowly and then retracts tightly against the perineum. Routine gentle traction fails.

1. What is happening, and what must you NOT do?

> **Shoulder dystocia**: the anterior shoulder is impacted behind the symphysis. **Do not** apply fundal pressure, pull hard, pull laterally, or let the mother keep pushing. **Note the time** of head delivery.

2. Outline your first-line actions.

> **HELPERR**:
>
> 1. **Call for help** (midwives, obstetrician, paediatrician). Stop pushing.
> 2. **Evaluate for episiotomy** if you need room for your hand.
> 3. **McRoberts**: lay her flat and hyperflex her thighs onto her abdomen (two assistants).
> 4. **Suprapubic pressure** from the side of the fetal back, pushing the anterior shoulder towards the fetal chest (continuous, then rocking), with gentle axial traction.

3. These fail. What next?

> **Internal manoeuvres**:
>
> - Rubin II: press behind the anterior shoulder towards the chest.
> - Woods' screw: press on the front of the posterior shoulder.
> - Reverse Woods.
>
> Or **deliver the posterior arm**: flex it at the elbow and sweep it across the chest. If these fail, move her **onto all fours** and repeat. Salvage options (PAST, clavicle fracture, Zavanelli) are last resorts.

4. The baby is delivered by posterior arm extraction. The right arm hangs limp and internally rotated, with the wrist flexed. What is it and what is the prognosis?

> **Erb's palsy** (C5–C6 brachial plexus injury): the 'waiter's tip' posture. Most recover within weeks to months. Refer for physiotherapy and review. Also check for a fractured clavicle or humerus and for HIE.

5. What maternal complications and documentation follow?

> - **PPH** (atony, trauma).
> - **Third- or fourth-degree tears**: examine carefully.
> - Bladder injury.
> - Debrief the family.
>
> Document: time of head delivery, time of body delivery, manoeuvres in order and who performed them, which shoulder was anterior, cord gases, and the neonatal condition. Plan counselling for the next pregnancy (recurrence risk; consider elective CS).

Take-away: Shoulder dystocia is a drill: help, McRoberts plus suprapubic pressure, then internal rotation or the posterior arm, then all-fours. Never use fundal pressure. Document every step with times.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.